

Matrices in Geometry - 1.9.6

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Problem Statement

If $\mathbf{Q}(0, 1)$ is equidistant from $\mathbf{P}(5, -3)$ and $\mathbf{R}(x, 6)$, find the value of x .

Solution

Given: $\mathbf{Q} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, $\mathbf{P} \begin{pmatrix} 5 \\ -3 \end{pmatrix}$ and a point $\mathbf{R} \begin{pmatrix} x \\ 6 \end{pmatrix}$ such that $\mathbf{Q}$ is equidistant from $\mathbf{P}$ and $\mathbf{R}$.

$$\therefore \|\mathbf{Q} - \mathbf{P}\| = \|\mathbf{Q} - \mathbf{R}\| \quad (1)$$

$$\text{On squaring both the sides, we get} \quad (2)$$

$$\|\mathbf{Q} - \mathbf{P}\|^2 = \|\mathbf{Q} - \mathbf{R}\|^2 \quad (3)$$

$$(\mathbf{Q} - \mathbf{P})^\top (\mathbf{Q} - \mathbf{P}) = (\mathbf{Q} - \mathbf{R})^\top (\mathbf{Q} - \mathbf{R}) \quad (4)$$

Solution

$$\mathbf{Q}^T \mathbf{Q} - 2\mathbf{Q}^T \mathbf{P} + \mathbf{P}^T \mathbf{P} = \mathbf{Q}^T \mathbf{Q} - 2\mathbf{Q}^T \mathbf{R} + \mathbf{R}^T \mathbf{R} \quad (5)$$

$$\|\mathbf{P}\|^2 - \|\mathbf{R}\|^2 = 2\mathbf{Q}^T (\mathbf{P} - \mathbf{R}) \quad (6)$$

$$\left\| \begin{pmatrix} 5 \\ -3 \end{pmatrix} \right\|^2 - \left\| \begin{pmatrix} x \\ 6 \end{pmatrix} \right\|^2 = 2 \begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} 5 - x \\ -9 \end{pmatrix} \quad (7)$$

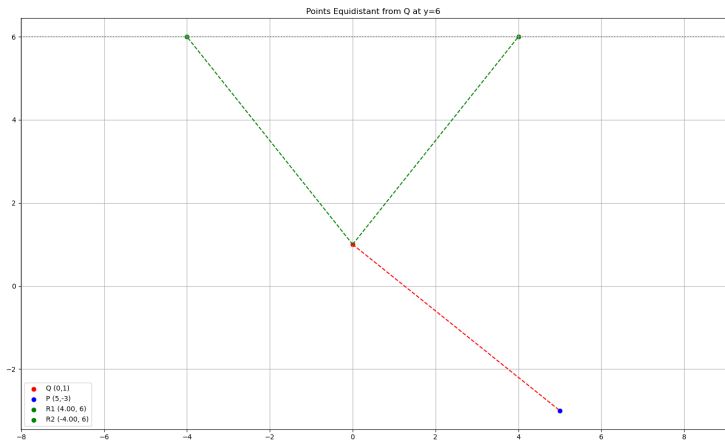
$$25 + 9 - x^2 - 36 = 2(0 + (-9)) \quad (8)$$

$$-2 - x^2 = 2(-9) \implies x^2 = 16 \implies x = -4, x = 4 \quad (9)$$

Final Answer

Hence, the final answer is $x = -4, x = 4$

(10)



C Code - Function to Find distance between points

```
#include <math.h>

// Array to store roots
static float roots[2];

// Function to return pointer to roots array
float* get_roots() {
    // distance between Q(0,1) and P(5,-3)
    float d1 = sqrt((5-0)*(5-0) + (-3-1)*(-3-1)); // s

    // Solve for x:  $x^2 + 25 = d1^2 \rightarrow x^2 = 16$ 
    roots[0] = 4.0;
    roots[1] = -4.0;

    return roots;
}
```

Python Code - for plotting

```
import matplotlib.pyplot as plt
import numpy as np

# Fixed points Q and P
Q = np.array([0, 1])
P = np.array([5, -3])

# Calculate distance between Q and P
d = np.linalg.norm(P - Q)

# y-coordinate of R
y_R = 6
```

Python Code - for plotting

```
# Find x-coordinates of points R on line y=6 such  
that distance(Q, R) = distance(Q, P)
```

```
x_R1 = np.sqrt(d**2 - 25)
```

```
x_R2 = -x_R1
```

```
R1 = np.array([x_R1, y_R])
```

```
R2 = np.array([x_R2, y_R])
```

```
# Plot points
```

```
plt.scatter(*Q, color='red', label='Q (0,1)')
```

```
plt.scatter(*P, color='blue', label='P (5,-3)')
```

```
plt.scatter(*R1, color='green', label=f'R1({x_R1:.2f},6)')
```

```
plt.scatter(*R2, color='green', label=f'R2({x_R2:.2f},6)')
```


Python Code - for plotting

```
# Plot lines from Q to P and Q to R points
plt.plot([Q[0], P[0]], [Q[1], P[1]], 'r--')
plt.plot([Q[0], R1[0]], [Q[1], R1[1]], 'g--')
plt.plot([Q[0], R2[0]], [Q[1], R2[1]], 'g--')

plt.axhline(y=y_R, color='gray', linestyle=':')

plt.legend()
plt.title('Points Equidistant from Q at y=6')
plt.grid(True)
plt.axis('equal')
plt.show()
```